

$$x(t) = \frac{1}{2} + \frac{2}{\pi}\left[\cos(\omega_0 t) - \frac{1}{3}\cos(3\omega_0 t) + \frac{1}{5}\cos(5\omega_0 t) - \frac{1}{7}\cos(7\omega_0 t) + \dots\right]$$

$$x_0(t) = 2x(t) - 1 = \frac{4}{\pi}\left[\quad\downarrow\quad\right]$$

$$x(t) = \sum_{n=-\infty}^{+\infty} x_n e^{jn\omega_0 t}$$

$$x_n = \frac{1}{T_0}\int_{T_0} x(t)\, e^{-jn\omega_0 t}\, dt$$

$$P_x = \sum_{n=-\infty}^{+\infty} |a_n|^2$$

Örnek: $x(t) = 2\sin(100t)$ işaretinin gücünü bulunuz

$$\omega_0 = 100 = \frac{2\pi}{T_0} \rightarrow T_0 = \frac{2\pi}{\omega_0} = \frac{2\pi}{100} = \frac{\pi}{50}$$

$$P_x = \frac{1}{T_0}\int_{T_0} |x(t)|^2 dt = \frac{1}{\pi/50}\int_{-\pi/100}^{\pi/100} 4\underbrace{\sin^2(100t)}_{\frac{1}{2}[1-\cos(200t)]}\, dt = \frac{100}{\pi}\left[t - \frac{\sin(200t)}{200}\right]_{-\pi/100}^{\pi/100}$$

$$= \frac{100}{\pi}\left\{\left[\frac{\pi}{100} - \frac{\sin\left(200\cdot\frac{\pi}{100}\right)}{200}\right] - \left[-\frac{\pi}{100} - \frac{\sin\left(200\cdot\frac{-\pi}{100}\right)}{2000}\right]\right\}$$

$$= \frac{100}{1}\cdot\frac{2\pi}{100} = 2\text{ watt}$$

$$P_x = \frac{A^2}{2} = \frac{2^2}{2} = 2\text{ watt}//$$

$$x(t) = 2\sin(100t) = \frac{e^{j100t} - e^{-j100t}}{2j} = \underbrace{\frac{1}{j}}_{x_1} e^{j100t} \underbrace{-\frac{1}{j}}_{x_{-1}} e^{-j100t}$$

$$P_x = \sum_n |x_n|^2 = \left|\frac{1}{j}\right|^2 + \left|-\frac{1}{j}\right|^2 = 1 + 1 = 2\text{ watt}$$

★ $\sin\alpha = \frac{e^{j\alpha} - e^{-j\alpha}}{2j}$ $\quad \cos\alpha = \frac{e^{j\alpha} + e^{-j\alpha}}{2}$

Fourier Dönüşümü

- Periyodik olmayan işaretler için
- Dirichlet şartları (To gerine belli bir aralıkta ifadesi g.)

$$X(f) = F\{x(t)\} = \int_{-\infty}^{+\infty} x(t)\, e^{-j2\pi ft}\, dt$$

$$x(t) = F^{-1}\{X(f)\} = \int_{-\infty}^{\infty} X(f)\cdot e^{j2\pi ft}\, df$$

$$x(t) \overset{F}{\longleftrightarrow} X(f) \Rightarrow X(f) = \underbrace{|X(f)|}_{\text{Genlik spektrum (magnitude) her zaman pozitif}} e^{j\theta_x(f)} \quad (\theta_x(f): \text{faz spektrumu})$$

Örnek : $x(t) = \delta(t)$

$$X(f) = \int_{-\infty}^{+\infty} \delta(t)\, e^{-j2\pi ft}\, dt = e^{-j2\pi f\cdot 0} = 1$$

A $\boxed{\delta(t) \overset{F}{\hookrightarrow} 1}$

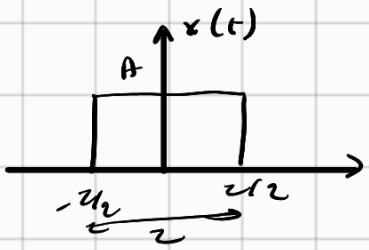
Örnek: $X(f) = \delta(f)$

$$x(t) = \int_{-\infty}^{\infty} \delta(f)\, e^{j2\pi ft}\, df = e^{j2\pi\cdot 0} = 1$$

$\boxed{1 \overset{F}{\hookrightarrow} \delta(f)}$

Örnek

$$x(t) = A\,\Pi\left(\frac{t}{\tau}\right)$$



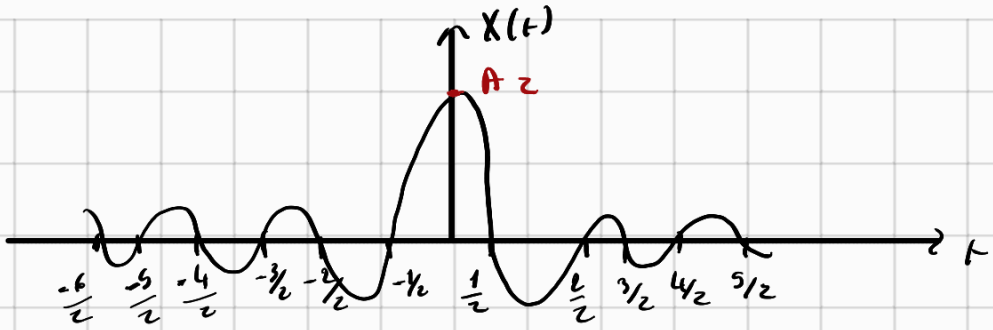
$$X(f) = \int_{-\tau/2}^{\tau/2} A\, e^{-j2\pi ft}\, dt$$

$$= A\,\frac{e^{-j2\pi ft}}{-j2\pi f}\Bigg|_{-\tau/2}^{\tau/2} = A\cdot\left[\frac{e^{-j2\pi f\tau/2} - e^{j2\pi f\tau/2}}{-j2\pi f}\right] =$$

$$X(f) = \frac{A}{\pi f\cdot\tau/2}\sin(\pi f\tau/2)\cdot\tau/2$$

$$X(f) = A\tau\,\mathrm{sinc}(f\tau)$$

$\boxed{x(t) = A\,\Pi\left(\frac{t}{\tau}\right) \overset{F}{\hookrightarrow} X(f) = A\tau\,\mathrm{sinc}(f\tau)}$



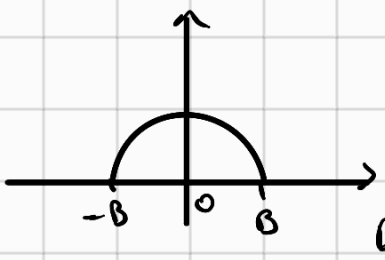
$f: z=k \in Z-\{0\} = \sin(k)=0$

$f = \frac{k}{2}$

Band Genişliği (B G)

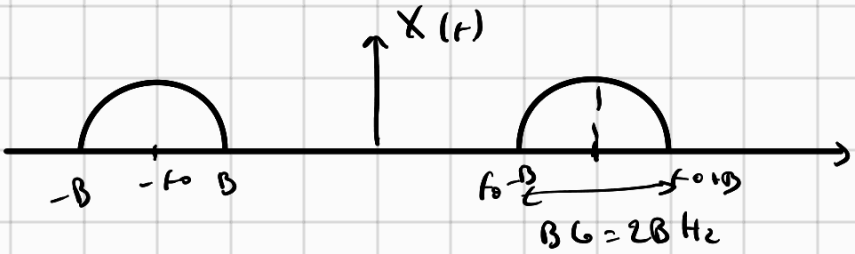
(Bandwidth-BW)

Temel bant işaretleri için (bileşenler 0Hz'in civarında) genlik spektrumun sıfırdan farklı olduğu en yüksek pozitif frekansa BG denir.



$BG = BHz$

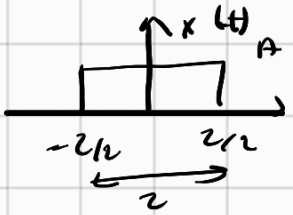
Band geçiren (Bandpass) işaretler için genlik spektrumun sıfırdan farklı olduğu pozitif frekanslar bölgesinin genişliğine BG denir



$BG = 2B\,Hz$

Sinc'in Teorikolarak $BG = \infty$

%90 BG ≡ effective bant genişliği ≅ BG = 1/2



Genişlik (zaman domaninde) 2 $\times$ BG (frekans domeninde) $BG = 1/2$

Veri hızıyla bant genişliği arasında doğrudan ilişkivar

FTM

Lineerlik özelliği :

$x_1(t) \xleftrightarrow{F} X_1(f)$

$x_2(t) \xleftrightarrow{F} X_2(f)$

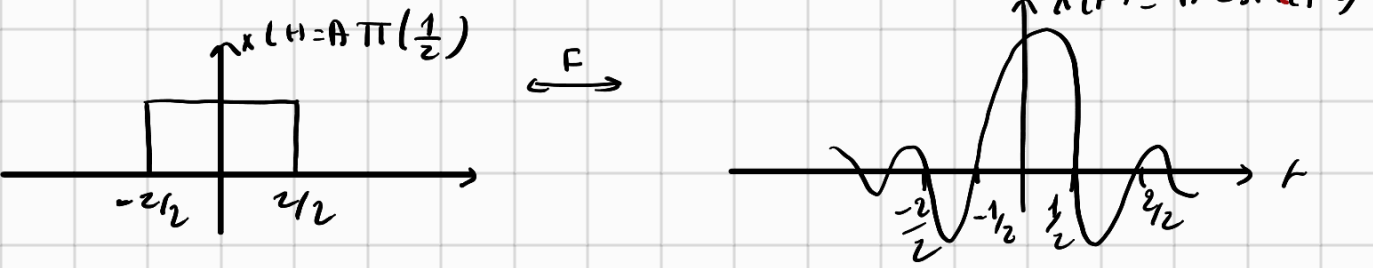
a, b, sabit

$x_3(t) = a\,x_1(t) + b\,x_2(t) \xleftrightarrow{F} a\,X_1(f) + b\,X_2(f) = X_3(f)$

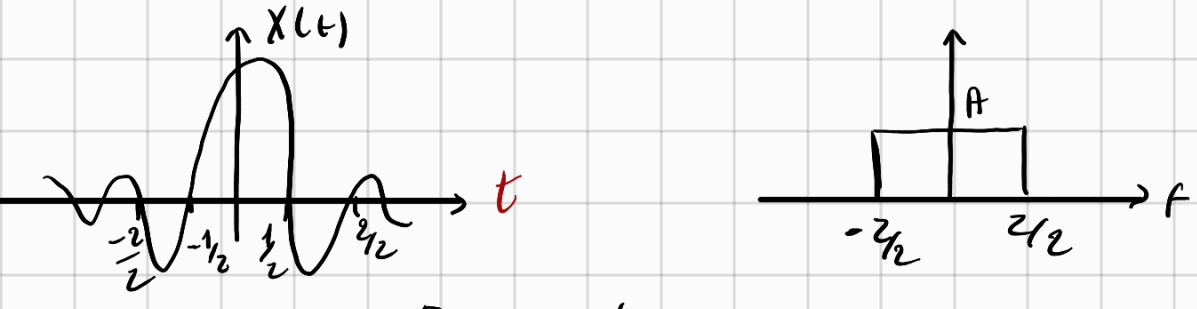
Dualite (Simetri) özelliği

$x(t) \xleftrightarrow{F} X(f)$

$X(t) \xleftrightarrow{F} x(-f)$



$x(t) = A\,2\,\mathrm{sinc}(t2) \xleftrightarrow{F} x(-f) = A\,\Pi\left(\frac{-f}{2}\right) = A\,\Pi\left(\frac{f}{2}\right)$



$A\,2\,\mathrm{sinc}(t2) \xleftrightarrow{F} A\,\Pi\left(\frac{f}{2}\right)$

Zamanda öteleme özelliği (Time shifting property)

$x(t) \xleftrightarrow{F} X(f)$

$x(t-t_0) \xleftrightarrow{F} X(f) \cdot e^{-j2\pi f t_0}$

İspat:

$$F\{x(t-t_0)\} = \int_{-\infty}^{\infty} x(t-t_0)\, e^{-j2\pi f t}\, dt$$

$t - t_0 = t' \Rightarrow t = t_0 + t'$

$dt = dt'$

$$= \int_{-\infty}^{+\infty} x(t')\, \underbrace{e^{-j2\pi f(t_0+t')}}_{e^{-j2\pi f t_0}\cdot e^{-j2\pi f t'}}\, dt'$$

$$= e^{-j2\pi f t_0} \underbrace{\int_{-\infty}^{\infty} x(t') e^{-j2\pi f t'} dt}_{X(f)}$$

Frekans ötelene özelliği

$$x(t) \xleftrightarrow{F} X(f)$$

$$x(t) \cdot e^{j2\pi f_0 t} \xleftrightarrow{F} X(f - f_0)$$

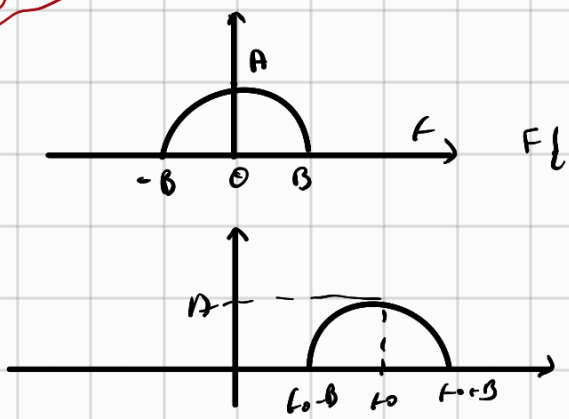
f_0 = sabit

İspat:

$$\mathcal{F}\{x(t) e^{j2\pi f_0 t}\} = \int_{-\infty}^{\infty} x(t) \underbrace{e^{j2\pi f_0 t} \cdot e^{-j2\pi f t}}_{e^{-j2\pi(f-f_0)t}} dt = \int_{-\infty}^{\infty} x(t) e^{-j2\pi \underbrace{(f-f_0)}_{f'} t} dt$$

$$= X(f') = X(f - f_0)$$

Örnek



$$\mathcal{F}\{x(t) \cdot e^{j2\pi f_0 t}\} = X(f - f_0)$$

Örnek: $x(t) = \cos(2\pi f_0 t) = \frac{1}{2}\left[e^{j2\pi f_0 t} + e^{-j2\pi f_0 t}\right]$

$1 \longleftrightarrow \delta(t)$

$x(t) e^{j2\pi f_0 t} \xleftrightarrow{F} X(f - f_0)$

1. $e^{j2\pi f_0 t} \xleftrightarrow{F} \delta(f - f_0)$ $\longrightarrow$ $x(t) = \cos(2\pi f_0 t)$

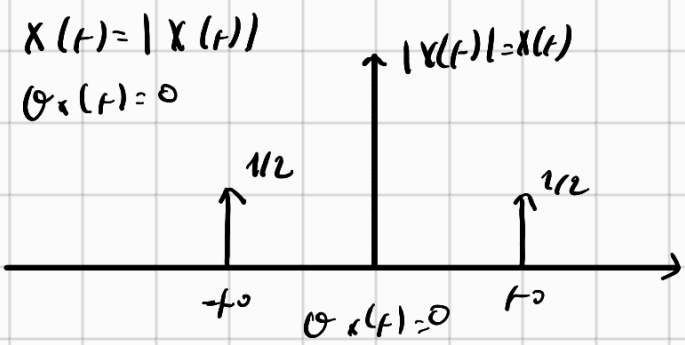
$\updownarrow F$

$X(f) = \frac{1}{2}\left[\delta(f - f_0) + \delta(f + f_0)\right]$

$$\cos(2\pi f_0 t) \xleftrightarrow{F} \frac{1}{2}\left[\delta(f - f_0) + \delta(f + f_0)\right]$$

$X(f) = |X(f)|$

$\theta_x(f) = 0$



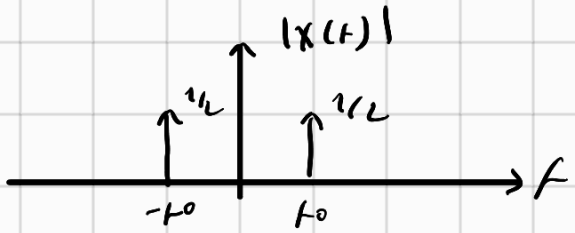
$$x(t) = \sin(2\pi f_0 t) = \frac{1}{2j}\left[e^{j2\pi f_0 t} - e^{-j2\pi f_0 t}\right]$$

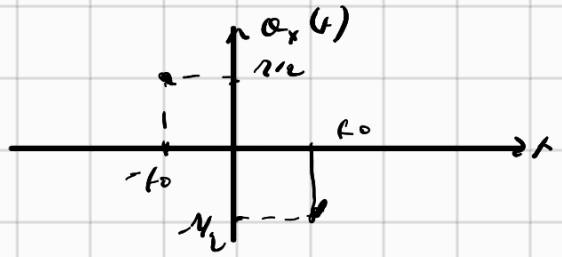
$$X(f) = \frac{1}{2j}\left[\delta(f-f_0) - \delta(f+f_0)\right]$$

$$j = e^{j\frac{\pi}{2}}$$

$e^{\mp ja} = \cos a \mp j\sin a$

$$X(f) = \frac{1}{2}\left[\delta(f-f_0)e^{-j\frac{\pi}{2}} + \delta(f+f_0)e^{j\frac{\pi}{2}}\right]$$





Modülasyon özelliği

$x(t) \xleftrightarrow{F} X(f)$

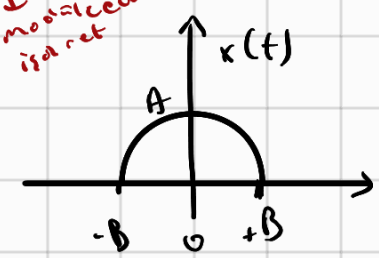
taşıyıcı işareti (carrier signal)

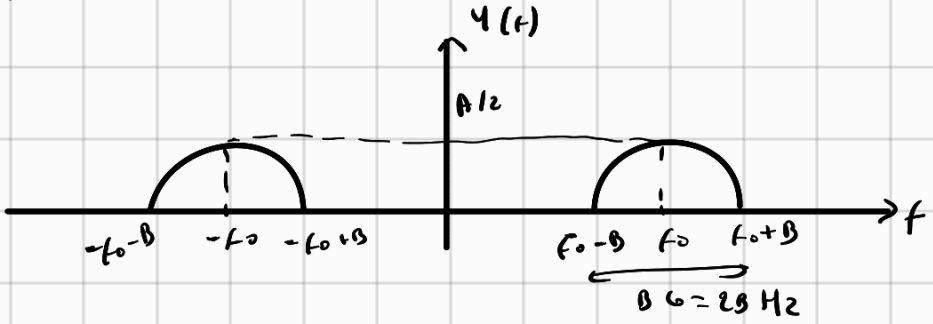
$$y(t) = x(t)\cdot\cos(2\pi f_0 t) \xleftrightarrow{F} Y(f) = \frac{1}{2}\left[X(f-f_0) + X(f+f_0)\right]$$

mesaj işareti

modüle edilmiş işaret

$$\cos(2\pi f_0 t) = \frac{1}{2}\left[e^{j2\pi f_0 t} + e^{-j2\pi f_0 t}\right]$$





frekans bölmeli gösterim

Ölçekleme özelliği (Scaling property)

$x(t) \xleftrightarrow{F} X(f)$

$$x(at) \xleftrightarrow{F} \frac{1}{|a|}X\left(\frac{f}{a}\right)$$

$a = 2$ $\quad x(2t) \leftrightarrow \frac{1}{2}X\left(\frac{f}{2}\right)$

$x(2t)$: zamanda daralmış

$\frac{1}{2}X\left(\frac{f}{2}\right)$: Frekansta genişlemiş

İşaretin Genişliği $\times$ BG

Konvülasyon özelliği:

$$x(t) \circledast y(t) = \int_{-\infty}^{\infty} x(z)\, y(t-z)\, dz$$

$$z(t) = x(t) \circledast y(t) \xleftrightarrow{F} Z(f) = X(f) \cdot Y(f)$$

$$z(t) = x(t) \cdot y(t) \xleftrightarrow{F} Z(f) = X(f) \circledast Y(f)$$

$$z(t) = x(t) \circledast y(t) \rightarrow \text{Genişlik}\{z(t)\} = \text{Genişlik}\{x(t)\} + \text{Genişlik}\{y(t)\}$$

$B_G = B_1$ $\quad$ $B_G = B_2$

$$z(t) = x(t) \cdot y(t) \Rightarrow Z(f) = X(f) \circledast Y(f)$$

$B_G = B_1$ $\quad$ $B_G = B_2$

$$B_G = B_1 + B_2$$

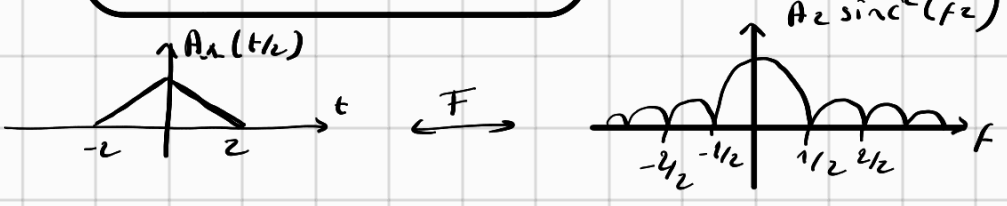
Örnek:

$$\Lambda(t) = \Pi(t) \circledast \Pi(t) \Rightarrow F\{A\Lambda(\tfrac{t}{z})\} = ?$$

$$F\{\Lambda(t)\} = \underbrace{F\{\Pi(t)\}}_{\text{sinc } f} \cdot \underbrace{F\{\Pi(t)\}}_{\text{sinc } f} = \text{sinc}^2(f)$$

$$F\{A\Lambda(t/z)\} = Az\,\text{sinc}^2(fz)$$

$$A\Lambda(t/z) \xleftrightarrow{F} Az\,\text{sinc}^2(fz)$$



İntegral özelliği

$x(t) \xleftrightarrow{F} X(f)$

$$y(t) = \int_{-\infty}^{t} x(z)\,dz \xleftrightarrow{F} Y(f) = \frac{X(f)}{j2\pi f} + \frac{1}{2}X(0)\delta(f)$$

dc bileşeni $= X(f)|_{f=0}$

frekans arttıkça genliği azalıyor
(alçak geçiren filtre gibi davranıyor)

Örnek: $u(t) = \int_{-\infty}^{t} \delta(z)\,dz \xleftrightarrow{F} \frac{1}{j2\pi f} + \frac{1}{2}\cdot 1\cdot\delta(f)$

$$u(t) \xleftrightarrow{F} \frac{1}{j2\pi f} + \frac{1}{2}\delta(f)$$

$$F\{u(t)\} = \frac{1}{j2\pi f} + \frac{1}{2}\delta(f)$$

bu $U(f)$ eşitliği

↳ bu frekans uzayında birim basamak

Örnek: $sgn(t) = u(t) - u(-t)$ ($\to a=-1$)

$$\mathcal{F}\{sgn(t)\} = \frac{1}{j2\pi f} + \frac{1}{2}\delta(f) - \frac{1}{|-1|}\left\{\frac{1}{j2\pi\frac{f}{-1}} + \frac{1}{2}\underbrace{\delta\left(\frac{f}{-1}\right)}_{=\delta(f)}\right\}$$

$$= \frac{1}{j\pi f}$$

$$\boxed{sgn(t) \xleftrightarrow{F} \frac{1}{j\pi f}}$$

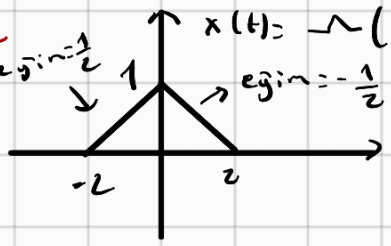
Türev özelliği

$x(t) \xleftrightarrow{F} X(f)$

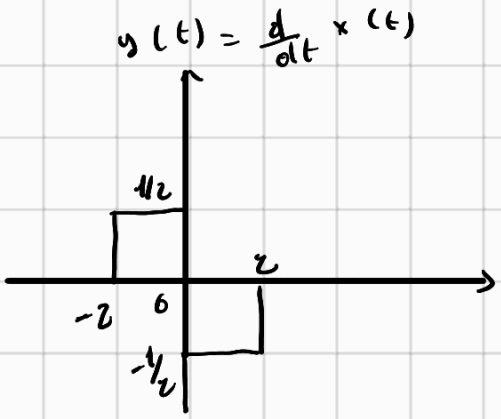
$y(t) = \frac{d}{dt}x(t) \xleftrightarrow{F} Y(f) = j2\pi f X(f)$ → frekans arttıkça işaret güçlenir (yüksek geçiren F)

$y(t) = \frac{d^n}{dt^n}x(t) \xleftrightarrow{F} Y(f) = (j2\pi f)^n X(f)$

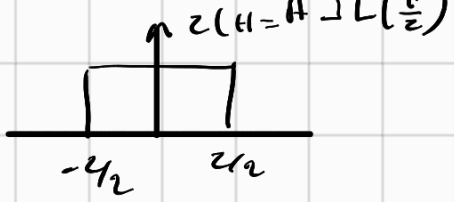
Örnek



$x(t) = \Lambda\left(\frac{t}{2}\right)$'nin F'ini türev özelliğini kullanarak bulunuz



$$y(t) = \frac{d}{dt} x(t)$$



$$Y(f) = j2\pi f X(f) \rightarrow X(f) = \frac{Y(f)}{j2\pi f}$$

$$y(t) = z(t+1) - z(t-1)$$

$$Y(f) = Z(f) \cdot e^{j2\pi f 1} - Z(f) e^{-j2\pi f 1}$$

$$Y(f) = Z(f) \left[e^{j2\pi f} - e^{-j2\pi f} \right]$$

$$X(f) = \frac{Y(f)}{j2\pi f} = \frac{Z(f)}{j2\pi f} \left[e^{j2\pi f} - e^{-j2\pi f} \right]$$

$$= \frac{Z(f)}{\pi f} \sin(2\pi f) \cdot 1$$

$$X(f) = Z(f) \cdot 2\,\mathrm{sinc}(2f) = 2\,\mathrm{sinc}^2(2f)$$

$$\downarrow$$

$$A\tau \sin(t\tau)$$

$$\downarrow$$

$$\tau/2$$

Periyodik işaretlerin FD

$\forall t,\ x(t+T_0) = x(t)$

$$x(t) = \sum_{n=-\infty}^{+\infty} x_n e^{jn\omega_0 t}$$

$$\omega_0 = 2\pi f_0 = \frac{2\pi}{T_0}$$

$$x_n = \frac{1}{T_0} \int_{T_0} x(t) e^{-jn\omega_0 t}$$

$$e^{j2\pi f_0 t} \xleftrightarrow{\mathcal{F}^{-1}} \delta(f - f_0)$$

$$e^{jn\omega_0 t} = e^{jn2\pi f_0 t} \xleftrightarrow{\mathcal{F}} \delta(f - nf_0)$$

$$x(t)=\sum_{n=-\infty}^{+\infty} x_n e^{j2\pi n f_0 t} \xleftrightarrow{\mathcal{F}} X(f)=\sum_{n=-\infty}^{+\infty} x_n \delta(f-nf_0)$$

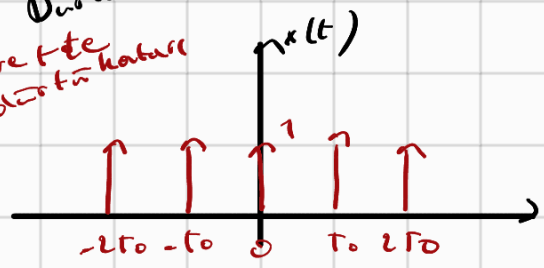
cosvesin fourier dönüşümünü içeriyor onun için mantıklı.

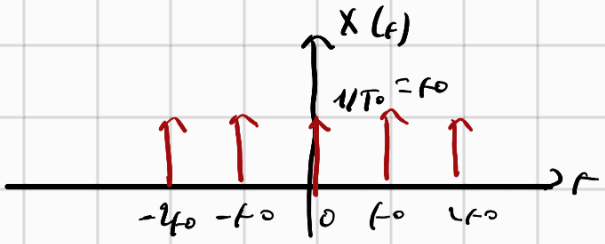
Örnek:

$$x(t)=\delta_{T_0}(t)=\sum_{n=-\infty}^{+\infty}\delta(t-nT_0)=\frac{1}{T_0}\sum_{-\infty}^{+\infty} e^{j2\pi n f_0 t} \xleftrightarrow{F} X(f)=\frac{1}{T_0}\sum_{-\infty}^{\infty}\delta(f-nf_0)$$

Dürtü katarı

Aşağıdaki dürtü katarı





Parseval Teoremi:

$$E_x=\int_{-\infty}^{+\infty}|x(t)|^2dt=\int_{-\infty}^{\infty}|X(f)|^2df$$

Sınav sorusu

Enerji yoğunluk fonksiyonu

A sin(t) enerjisini bulun ama fourier dönüşümünden gidilecek.

$x(t)$ reel $\Rightarrow X(-f)=X^*(f)$

$x(t)$ reel ve çift $\Rightarrow X(f)$ reel ve çift $\rightarrow X(-f)=X(f)$ + ödev ispatla

$x^*(t)=x(t)$ $x(-t)=x(t)$ $\hookrightarrow X^*(f)=X(f)$ ✓

$x(t)$ sanal ve tekse $X(f)$ sanal ve tek

$\downarrow$ $\hookrightarrow X(-f)=-X(f)$

$X^*(f)=-X(f)$